RIRPA in Vayesta

Current state, cost, and a plan for HHT/CDERI improvements

10 November 2026

Nothing in this note was run. All scaling statements are operation counts. The measured numbers in Section 3.1 are quoted from the `monomial_GW` and `Davidson-gw` records, and the source is named each time.

Symbol	Meaning
o, v	occupied / virtual orbitals of the full system
ov	number of spatial particle–hole pairs, $o\,v$ (spin-orbital codes use $2ov$)
N_{aux}	number of auxiliary (density-fitting) functions; $\mathcal{O}(N)$
L	occupied–virtual CDERIs, shape (N_{aux}, ov)
D	diagonal of orbital-energy differences $\epsilon_a - \epsilon_i$, length ov
n_t	number of rows of `target_rot`, the space the moment is projected onto
N_p	number of quadrature points (Clenshaw–Curtis) or poles (HHT)
N_{frag}	number of fragments in an embedding calculation

1 Vayesta as a whole

Vayesta is a Python quantum-embedding package built on PySCF. It partitions a molecule or solid into fragments, builds a cluster for each fragment (fragment orbitals plus a bath), solves each cluster with a correlated solver (CCSD, FCI, MP2, EBFCI, ...), and assembles global properties from the fragment results. The main subpackages are:

- `core/`: fragmentation, bath construction (`core/bath/`: DMET, MP2 and RPA bath natural orbitals, bosonic baths), integrals (`core/eris.py`, where `get_cderi` lives), screening (`core/screening/`), and the generic embedding driver `core/qemb/`.
- `ewf/`, `dmet/`, `edmet/`: the embedding methods themselves (wave-function embedding, DMET, and extended/bosonic DMET).
- `solver/`: cluster solvers, including the electron–boson ones that EDMET needs.
- `rpa/`: full-system RPA. `rpa.py`, `ssrpa.py` and `ssurpa.py` are dense $\mathcal{O}(N^6)$ reference implementations. `rirpa/` is the density-fitted, moment-based code that the embedding layers use in production.

RIRPA matters because it supplies *global long-range information to local clusters*. It is called from four places:

Caller	What it asks RIRPA for	`target_rot`
`core/screening/screening_moment.py` (`mrpa` screening, cRPA)	zeroth dd moment, then only $T_f\,\eta_0\,T_f^{\mathsf{T}}$ per fragment	stacked cluster ph spaces, spin-orbital, $n_t = \sum_f ov_f^{\mathrm{clus}}$
`core/bath/rpa.py` (RPA bath NOs)	zeroth moment, then $m_0 T^{\mathsf{T}}$	$R_o \otimes R_v$: $o\,v_{\mathrm{clus}}$ or $o_{\mathrm{clus}}\,v$ rows; **one RIRPA object per fragment**
`core/qemb/qemb.py` (bosonic bath; external dRPA correction)	zeroth moment / total energy	bosonic targets; or L itself (cumulant energy)
`edmet/edmet.py`	zeroth moment and energy **with an xc kernel** (`rixc`)	cluster ph and bosonic spaces

2 How `vayesta/rpa/rirpa` is implemented

2.1 Files and classes

File	Contents
`__init__.py`	`ssRIRPA(mf, ...)` factory: UHF goes to `ssRIURPA`, a non-`None` `rixc` goes to `ssRIRRPA`, and everything else goes to `ssRIdRRPA`.
`RIRPA.py`	`ssRIRRPA`: the general (xc-kernel-capable) restricted class, which works in the **spin-orbital** $2ov$ space; also `construct_product_RI`, `construct_inverse_RI` and `compress_low_rank`.
`RIdRRPA.py`	`ssRIdRRPA`: specialised direct-RPA RHF path in **spatial** ov with explicit spin factors (the path most embedding calls take).
`RIURPA.py`	`ssRIURPA`: unrestricted variant; overrides only the integral/xc construction.
`NI_eval.py`	Numerical-integration framework: Clenshaw–Curtis on infinite and semi-infinite ranges with a nested error estimate, Gauss–Laguerre, and the optimisation of the grid scale a.
`momzero_NI.py`	Integrands for the zeroth moment (`MomzeroDeductNone/D/HigherOrder`, plus the `_dRHF` variant) and the analytic-offset integrand (`MomzeroOffsetCalcGaussLag`).
`energy_NI.py`	Integrand for $\mathrm{Tr}(MP)^{1/2}$ (`NITrRootMP`, plus `_dRHF`).

2.2 The mathematics as coded

This follows Furche (PRB **64**, 195120) and arXiv:2301.09107. For restricted dRPA in spatial orbitals,

$$A - B = D \quad \text{(diagonal)}, \qquad A + B = D + 4L^{\mathsf{T}}L, \tag{1}$$

$$MP \equiv (A-B)(A+B) = D^2 + S_L^{\mathsf{T}} S_R \quad (\text{rank } N_{\mathrm{aux}}), \qquad \eta_0 = (MP)^{1/2}(A+B)^{-1}. \tag{2}$$

Vayesta stores $A + B$ as $(\sqrt{2}L)^{\mathsf{T}}(\sqrt{2}L)$ with a further factor of 2 for spin. The square root is evaluated as a frequency integral,

$$(MP)^{1/2} = \frac{1}{\pi}\int_{-\infty}^{\infty} \Big[\, I - \omega^2 (MP + \omega^2)^{-1} \Big]\, d\omega, \tag{3}$$

and Woodbury reduces the resolvent to auxiliary space. At each frequency,

$$F(\omega) = (D^2 + \omega^2)^{-1} \quad \text{(diagonal)}, \tag{4}$$

$$Q(\omega) = S_R\, F(\omega)\, S_L^{\mathsf{T}} \quad (N_{\mathrm{aux}} \times N_{\mathrm{aux}}) \qquad \leftarrow \text{ the } N_{\mathrm{aux}}^2 ov \text{ step}, \tag{5}$$

$$\mathrm{term}(\omega) = (TF)\, S_L^{\mathsf{T}} \big[(I+Q)^{-1} - I\big]\, S_R F \quad (n_t \times ov). \tag{6}$$

The default `"HO"` scheme subtracts the diagonal and the first-order term $-I$ inside the integrand and adds them back analytically. The first-order offset,

$$T\int_0^\infty e^{-\omega D}\, S_L^\mathsf{T} S_R\, e^{-\omega D}\, d\omega, \qquad \text{exactly} \quad \frac{(S_L^\mathsf{T} S_R)_{pq}}{D_p + D_q}, \tag{7}$$

would cost $\mathcal{O}(n_t\, ov^2)$ if done exactly. Vayesta therefore evaluates it with a **second, separate 48-point Gauss–Laguerre integral**, whose scale is also optimised.

The remaining steps of the zeroth moment are as follows.

1. Fit the grid scale a for each integral (`opt_quadrature_diag`). The fit uses an exponential search over 14 decades, then Newton or bounded minimisation, so that the rule integrates a diagonal approximation exactly. Each probe costs $\mathcal{O}(N_p\, ov)$.
2. Integrate with Clenshaw–Curtis (`npoints=48`, rounded up to a multiple of 4). The $\frac{1}{2}$- and $\frac{1}{4}$-grids are nested inside it, and they give the error estimate.
3. Form the explicit inverse `np.linalg.inv(I + Q)` at every point.
4. Multiply by $(A+B)^{-1}$, itself built as an RI correction (`construct_inverse_RI`: an $N_\text{aux} \times N_\text{aux}$ inverse plus an SVD; or, in the dRPA path, an explicit $\left(I + 2LD^{-1}L^\mathsf{T}\right)^{-1}$).
5. Optionally compress (`compress > 0`) by a non-symmetric `eig` of $S_L S_R^\mathsf{T}$ with an absolute 10^{-12} cut. This is off by default.

Higher moments use the exact recurrence $\eta_{n+2} = \eta_n D^2 + (\eta_n S_R^\mathsf{T}) S_L$. Each order is two GEMMs of $\mathcal{O}(n_t N_\text{aux} ov)$, which is already optimal.

The energy (`kernel_energy`) is $E_c = \frac{1}{2}\big[\operatorname{Tr}(MP)^{1/2} - \operatorname{Tr} A\big]$. The trace is another optimised Clenshaw–Curtis integral (`NITrRootMP`), and its diagonal and first-order parts are added analytically.

The xc-kernel path (`ssRIRRPA` with `rixc`, used by EDMET) differs in two ways. First, $A - B$ is no longer diagonal, so `construct_product_RI` builds the non-symmetric product with rank $N_\text{aux}(A+B) + N_\text{aux}(A-B)$, which is up to 2–3× the CDERI rank. Second, the whole calculation runs in the $2ov$ spin-orbital space.

`get_gap` runs a **non-symmetric** Davidson (`pyscf.lib.eig`, left and right vectors) on MP.

2.3 Current cost

Per call of `kernel_moms(0, target_rot)` on the dRPA RHF path (`ssRIdRRPA`), with $N_p = 48$:

Step	Cost	Notes
CDERI transform `get_cderi` (if `lov` not passed)	$\mathcal{O}(N_{\text{aux}}\, n_{\text{ao}}^2\, o)$	N^4; repeated by every caller that does not pass `lov`
Build S_L, S_R	$\mathcal{O}(N_{\text{aux}} ov)$	two to three $N_{\text{aux}} \times ov$ arrays held at once
Grid fit, both integrals	$\mathcal{O}(N_{\text{it}} N_p ov)$	cheap, but runs every call
Main integral: $Q(\omega)$	$N_p\, N_{\text{aux}}^2 ov$	**dominant for small** n_t
Main integral: $\text{inv}(I+Q)$	$N_p\, N_{\text{aux}}^3$	explicit inverse
Main integral: target contractions	$N_p(2n_t N_{\text{aux}} ov + n_t N_{\text{aux}}^2)$	**dominant for large** n_t
Offset integral (Gauss–Laguerre)	$N_p \cdot 2n_t N_{\text{aux}} ov$	a second pass, as expensive as the target part of the main integral
$(A+B)^{-1}$	$N_{\text{aux}}^2 ov + N_{\text{aux}}^3 + 2n_t N_{\text{aux}} ov$	needed only because of the $(MP)^{1/2}(A+B)^{-1}$ form
Spin stacking (spin-orbital targets)	$\mathcal{O}(n_t^2 ov + n_t^3)$	`compress_target_rot` forms an $n_t \times n_t$ Gram
Higher moments	$2n_t N_{\text{aux}} ov$ per order	fine

The leading terms are therefore

$$\text{cost}(\eta_0) \approx 48\, N_{\text{aux}}^2 ov + {\sim}190\, n_t N_{\text{aux}} ov + 48\, N_{\text{aux}}^3, \qquad \text{memory} \approx (2\text{–}3)\, N_{\text{aux}} ov + (2 + n_{\text{mom}})\, n_t ov. \tag{8}$$

This is $\mathcal{O}(N^4)$ when $n_t = \mathcal{O}(N)$, and $\mathcal{O}(N^5)$ for the full moment ($n_t = ov$, which the code warns about). The energy costs ${\sim}96\, N_{\text{aux}}^2 ov$ (Q plus the back contraction at every point). For the xc-kernel path, replace N_{aux} by the 2–3× larger product rank and ov by $2ov$.

How the call pattern turns this into total cost.

- **RPA bath:** a new `ssRIdRRPA` is built for every fragment. So the fragment-independent $48\, N_{\text{aux}}^2 ov$ work (the $Q(\omega)$ matrices and inverses) and the grid fit are repeated N_{frag} times. With a dense Kronecker target of $n_t \sim o\, v_{\text{clus}}$ rows, each fragment is $\mathcal{O}(N^4)$, and the calculation is $\mathcal{O}(N^5)$ overall.
- **Screening:** one call with all fragments stacked. It forms the full $n_t \times 2ov$ result and then keeps only the diagonal fragment blocks $T_f \eta_0 T_f^{\mathsf{T}}$, so most of the $n_t \times ov$ back-contraction is thrown away.
- **Accuracy:** the error is controlled only by the nested-Clenshaw–Curtis estimate and a warning above `err_tol`$= 10^{-6}$ per element. It is not certified. Section 3.1 gives `momentGW`'s measurement of this same route at about 10^{-9} relative error on a small-gap system, and an unfitted 48-point Clenshaw–Curtis rule measured 5×10^{-7} on a larger one.

3 Bringing in the monomial / Davidson improvements

3.1 What those projects established

Both `monomial_GW` (a momentGW fork whose legacy Clenshaw–Curtis η_0 route *is* this Vayesta route, Gauss–Laguerre offset included) and `Davidson-gw` (`gwfast`) replaced the frequency integral with the **Hale–Higham–Trefethen (HHT) / Zolotarev rational approximation of the inverse square root**. They write it in the symmetric, $(A+B)^{-1}$-free form

$$\eta_0 = D^{1/2}\, \tilde{M}^{-1/2}\, D^{1/2}, \qquad \tilde{M} = D^2 + 4WW^{\mathsf{T}}, \qquad W = D^{1/2} L^{\mathsf{T}} \quad \text{(symmetric positive definite)}, \tag{9}$$

$$x^{-1/2} \approx \sum_{j=1}^{N_p} \frac{w_j}{x + s_j} \qquad \text{on a certified } [\lambda_{\min}, \lambda_{\max}] \supseteq \operatorname{spec}(\tilde{M}). \tag{10}$$

Each pole is one resolvent, reduced exactly to auxiliary space by Woodbury:

$$F_j = D\,(D^2 + s_j)^{-1} \quad \text{(diagonal)}, \tag{11}$$

$$C_j = I + 4\,LF_jL^{\mathsf{T}} \quad (N_{\text{aux}} \times N_{\text{aux}},\ \text{SPD}) \qquad \leftarrow \text{ one Gram, } N_{\text{aux}}^2 ov, \tag{12}$$

$$\eta_0 \approx \sum_j w_j \Big[F_j - 4\,F_jL^{\mathsf{T}}C_j^{-1}LF_j\Big]. \tag{13}$$

The components, all implemented and tested in at least one of the two repositories:

1. **Certified spectral interval.**

   $$\lambda_{\min} = \min(D)^2, \qquad \lambda_{\max} = \max(D^2) + 4\min\left(\|W\|_F^2,\ \|W\|_1\|W\|_\infty\right).$$

   `gwfast` also has a tighter `"spectral"` option, which uses the top eigenvalue of one $N_{\text{aux}} \times N_{\text{aux}}$ Gram. Both ends are padded outwards.

2. **Pole count from a measured scalar error.** The rule is the smallest N_p for which $\max_x |1 - \sqrt{x}\,r(x)| \le \text{tol}$ over the interval. N_p grows only like $\log\kappa\,\log(1/\text{tol})$. The coefficients come from AGM and descending-Landen elliptic functions in extended precision, so $1 - k^2$ is never formed (`momentGW/eta0.py`, `gwfast/quadrature.py`).

3. **No grid fit and no offset integral.** The rational sum absorbs the bare and first-order terms: at zero coupling each solve is the identity. This removes Vayesta's `"HO"`/Gauss–Laguerre second pass and its a-optimisation.

4. **Cholesky solves with residual checks instead of `inv`.** `momentGW` measured that this is an accuracy rule rather than a speed rule, because both are dominated by the $N_{\text{aux}}^2 ov$ apply.

5. **No $ov \times ov$ intermediate.** The code records intermediate shapes and a test asserts that none is formed.

6. **Natural auxiliary functions** (`gwfast.integral.natural_auxiliary_functions`). These rotate the CDERIs to the eigenvectors of LL^{T} above a threshold. The rotation is symmetric (`eigh`) and does not change the result above that threshold. Compare Vayesta's off-by-default, non-symmetric `eig` compression.

7. **Square root and inverse square root together** (`gwfast.rpa.get_root_operators`). The expansion

   $$\tilde{M}^{1/2} \approx \sum_j w_j [I - s_j(\tilde{M} + s_j)^{-1}]$$

   reuses the same C_j, so one pass over the poles gives both.

Measured results, quoted from those projects:

Source	Comparison	Result
`monomial_GW/ROADMAP.md`, Milestone 2	ozone/PBE (condition 6.7×10^4) η_0 vs a dense oracle	Clenshaw–Curtis-48: 1.15×10^{-9}; HHT-26: 2.1×10^{-13}
same	benzene/cc-pVDZ η_0 stage	20 poles vs 48 points, **1.82**× faster
`Davidson-gw/docs/hht-quadrature-benchmark-2026-09-22.md`	C_8H_{18}/def2-TZVPP ($ov = 15411$, $N_{\text{aux}} = 1148$)	unfitted Clenshaw–Curtis-48: 5×10^{-7}; HHT-20: 3×10^{-14}; **1.95**× faster at 20 poles, 2.72× at 16; peak memory roughly halved

3.2 Applying it to Vayesta's projected moments

Vayesta needs $T\eta_0$ for a general target T ($n_t \times ov$), not just $\eta_0 L^\mathsf{T}$ as in GW. The Woodbury form still applies:

$$T\eta_0 \approx \underbrace{T\ \mathrm{diag}\Big(\sum_j w_j F_j\Big)}_{\approx T} - 4\sum_j w_j\ \underbrace{(TF_jL^\mathsf{T})}_{n_t N_{\mathrm{aux}} ov}\ \underbrace{C_j^{-1}}_{\text{solve}}\ \underbrace{(LF_j)}_{n_t N_{\mathrm{aux}} ov}. \tag{14}$$

Per pole this costs $N_{\mathrm{aux}}^2 ov$ (Gram) + $N_{\mathrm{aux}}^3/3$ (Cholesky) + $2n_t N_{\mathrm{aux}} ov + n_t N_{\mathrm{aux}}^2$. The scaling class is the same as today. The gain comes from having fewer passes and dropping the redundant ones:

	Current (CC-48, `"HO"`)	HHT, $N_p \approx 20$	Ratio
$N_{\mathrm{aux}}^2 ov$ work	48	20	2.4×
$n_t N_{\mathrm{aux}} ov$ work	∼96 (main) +∼96 (offset) +2	∼40	∼4.8×
N_{aux}^3	48 explicit inverses + SVD	20 Cholesky factorisations	∼7×
Grid fitting, $(A+B)^{-1}$	yes	not needed	–
Error control	heuristic, about 10^{-7} to 10^{-9}	certified to tol (10^{-14} default in `momentGW`)	–

The expected η_0 speed-up is therefore about **2.4× when** $n_t \ll N_{\mathrm{aux}}$**, about 4× when** $n_t \sim N_{\mathrm{aux}}$**, and about 4.8× when** $n_t \gg N_{\mathrm{aux}}$, with accuracy improving by 5 to 7 orders of magnitude. These are operation-count estimates and must be measured. The one directly comparable measurement is 1.82× on the GW η_0 stage, where $n_t = N_{\mathrm{aux}}$.
The same pole factors give further savings that depend on what each caller actually needs:

1. **Cache the per-pole factors across targets and fragments.** F_j, C_j and its Cholesky factor depend only on the mean field, not on T. Build them once per calculation, at $N_p N_{\mathrm{aux}}^2 ov$ and $N_p N_{\mathrm{aux}}^2$ memory, and pass them to every caller. For the **RPA bath** this changes the fragment-independent part from $N_{\mathrm{frag}} \cdot 48\, N_{\mathrm{aux}}^2 ov$ to $20\, N_{\mathrm{aux}}^2 ov$ once. That is a reduction of $\sim 2.4\, N_{\mathrm{frag}}\times$ on that term, and it changes the overall scaling from $\mathcal{O}(N^5)$ to $\mathcal{O}(N^4)$ plus a per-fragment target term.

2. **Two-sided projections.** Several callers contract the ov index of η_0 on both sides, and the left and right targets need not be the same, or even have the same number of rows. With a left target X ($n_X \times ov$) and a right target Y ($n_Y \times ov$), η_0 is symmetric, so

$$\begin{aligned} X\eta_0 Y^\mathsf{T} &\approx X\ \mathrm{diag}\Big(\sum_j w_j F_j\Big) Y^\mathsf{T} - 4\sum_j w_j\, B_j^X C_j^{-1} (B_j^Y)^\mathsf{T}, \\ B_j^X &= XF_jL^\mathsf{T}\ (n_X \times N_{\mathrm{aux}}), \qquad B_j^Y = YF_jL^\mathsf{T}\ (n_Y \times N_{\mathrm{aux}}). \end{aligned} \tag{15}$$

   The result carries only target labels ($n_X \times n_Y$), with no particle–hole index left. When the two target labels are contracted against each other it is a single number, labelled only by the pair (X, Y). That is a trace in auxiliary space:

$$\mathrm{Tr}\,[X\eta_0 Y^\mathsf{T}] \approx \mathrm{Tr}\Big[X\ \mathrm{diag}\Big(\sum_j w_j F_j\Big) Y^\mathsf{T}\Big] - 4\sum_j w_j\ \mathrm{Tr}\Big[C_j^{-1}(B_j^Y)^\mathsf{T} B_j^X\Big]. \tag{16}$$

   Neither form ever builds the $n_X \times ov$ intermediate $X\eta_0$. Per pole the cost is $(n_X + n_Y) N_{\mathrm{aux}} ov$ for the two B matrices, which drops to $n_X N_{\mathrm{aux}} ov$ when $Y = X$. Then the open form costs $n_X N_{\mathrm{aux}}^2 + n_X n_Y N_{\mathrm{aux}}$, and the trace costs $\min(n_X, n_Y) N_{\mathrm{aux}}^2 + N_{\mathrm{aux}}^3$ instead. Today the cost is $2n_X N_{\mathrm{aux}} ov$ (main integral) plus $2n_X N_{\mathrm{aux}} ov$ (offset) per point, and then a further $n_X n_Y ov$ for the projection.

   Where this applies in Vayesta:

Caller	X	Y	Form
`screening_moment.py` (`dot(momzero[..], rot.T)`)	T_f	T_f	open, diagonal blocks only
`bath/rpa.py` (`dot(m0, target_rot.T)`)	$R_o \otimes R_v$	the same	open
`qemb.py` cumulant energy (`m0 - l_` against `l_2`)	`l_`	`l_2`, which differs from `l_` by the sign of the negative-CDERI rows	trace
`RIRPA.kernel_energy`, `"linear"` xc correction	`ri_b_xc[0]`	`ri_b_xc[1]`	trace
`RIRPA.direct_AC_integration`	`lrot`	`rrot`	trace

When either side lies in the RI span the matching B is already known. For $Y = L$, $B_j^Y = LF_jL^\mathsf{T} = (C_j - I)/4$. The cumulant energy without negative CDERIs has $X = Y = L$, and its trace reduces to $-\frac{1}{4}\sum_j w_j \operatorname{Tr}\left[C_j - 2I + C_j^{-1}\right]$ plus the diagonal term. That is one Gram and one Cholesky factorisation per pole, with no target contraction at all.

Keep the rational diagonal $\sum_j w_j F_j$ rather than replacing it by I. The callers subtract the bare target (`m0 - l_`, `eta0 - lrot`), and the consistent diagonal makes that difference come out of the low-rank term alone instead of from a cancellation of two large numbers.

3. **Kronecker-structured targets.** The bath target is $R_o \otimes R_v$, but it is stored densely, and `screening_moment` does the same per spin. Applying it in factored form, $B_j[I, A, Q] = \sum_{ia} R_o[I, i]\, R_v[A, a]\, (F_jL^\mathsf{T})[ia, Q]$ done as two small GEMMs, saves a factor of about o (or v) per contraction. With item 1, the bath then costs about $\mathcal{O}(o\, v\, v_{\mathrm{clus}} N_{\mathrm{aux}})$ per fragment and pole, which is $\mathcal{O}(N^3)$ per fragment and $\mathcal{O}(N^4)$ overall.

4. **Targets inside the RI span.** For the cumulant energy in `qemb` the target is L itself, and TF_jL^T *is* the Gram, $(C_j - I)/4$. This is exactly the `momentGW` kernel, at $\sim 2N_{\mathrm{aux}}^2 ov$ per pole and no n_t term.

5. **Energy from the same poles.** Applying the rule to $\sqrt{x} = x \cdot x^{-1/2}$ for both $\tilde{M}$ and D^2, and differencing so that the leading terms cancel identically, gives

$$\operatorname{Tr}\left(\tilde{M}^{1/2}\right) - \operatorname{Tr}(D) \approx 4 \sum_j w_j s_j \operatorname{Tr}\Big[C_j^{-1}\big(L\, D\, (D^2 + s_j)^{-2} L^\mathsf{T}\big)\Big]. \tag{17}$$

This costs $\sim 2N_{\mathrm{aux}}^2 ov$ per pole, against $\sim 96\, N_{\mathrm{aux}}^2 ov$ for `NITrRootMP` today: about 2.4× faster and certified. One caveat: the bound is relative per eigenvalue, so the absolute energy error is at most about $2\,\mathrm{tol}\,\operatorname{Tr}(D)$. Choose tol against the required E_c accuracy. At 10^{-14} this is well under a microhartree for any system Vayesta handles.

6. **Natural auxiliary functions.** Rotating L once to $N'_{\mathrm{aux}} \le N_{\mathrm{aux}}$ NAFs saves $(N_{\mathrm{aux}}/N'_{\mathrm{aux}})^2$ on every Gram and $N_{\mathrm{aux}}/N'_{\mathrm{aux}}$ on every target contraction. The saving depends on the system and threshold and has not been measured here. The rotation must use a scale-aware PSD criterion and report the rank and discarded norm, which is `monomial_GW` Milestone 4's open item. The current absolute 10^{-12} cut does neither.

7. **Symmetric gap and eigenvectors.** $\tilde{M}$ is symmetric with eigenvalues Ω^2. `get_gap` can therefore use the symmetric `pyscf.lib.davidson` on $x \mapsto D^2 x + 4W(W^\mathsf{T} x)$ ($\mathcal{O}(N_{\mathrm{aux}} ov)$ per matvec) instead of the left/right non-symmetric `eig` on MP followed by re-biorthogonalisation.

Rough projected totals for the two main embedding uses (operation counts, before NAF):

Use	Today	With the changes	Gain
RPA bath, N_{frag} fragments	$N_{\mathrm{frag}}\big(48N_{\mathrm{aux}}^2ov + \sim 190\, o\, v_c N_{\mathrm{aux}} ov\big) \sim \mathcal{O}(N^5)$	$20N_{\mathrm{aux}}^2ov + N_{\mathrm{frag}} \cdot 20\big(o\, v\, v_c N_{\mathrm{aux}} + \dots\big) \sim \mathcal{O}(N^4)$	grows with system size; already about $N_{\mathrm{frag}} \times o$ on the target term
`mrpa` screening	$48N_{\mathrm{aux}}^2ov + \sim 190\, n_t N_{\mathrm{aux}} ov$, $n_t = \sum_f 2ov_f$	$20N_{\mathrm{aux}}^2ov + 20\sum_f \big(2n_f N_{\mathrm{aux}} ov/k + n_f N_{\mathrm{aux}}^2\big)$, where k is the Kronecker saving	about 5×, and more with Kronecker targets
Total dRPA energy	$\sim 96N_{\mathrm{aux}}^2ov$	$\sim 40N_{\mathrm{aux}}^2ov$	about 2.4×, certified

3.3 Proposed sequence

This mirrors `monomial_GW`'s ordering: restricted dRPA first, measured before it becomes the default, and the old route kept as an independent reference.

1. **Baseline.** Record η_0, the projected moments, energies, screened interactions and bath occupations for a small set (H_2O, a small-gap molecule, one Hubbard case, one solid), using the existing `vayesta/tests/rpa/test_rirpa_*` systems plus a dense `ssRPA` oracle.
2. **HHT kernel for `ssRIdRRPA`.** Port `momentGW/eta0.py` (interval, poles, scalar certificate) and a projected apply that accepts T in one-sided, two-sided and in-RI-span forms. Add an `eta0_method={"clencur","hht"}` option. Keep Clenshaw–Curtis as the default until the baseline agrees within the certified error, then switch.
3. **Pole-factor cache and caller changes.** Build the factors once per embedding object. Move `bath/rpa.py` and `screening_moment.py` to the two-sided, Kronecker-factored apply, and stop recomputing CDERIs per fragment by passing `lov` everywhere.
4. **Energy and gap** from the same poles (items 5 and 7 above).
5. **NAF/scale-aware compression** behind an option, reporting rank, discarded norm and the observed effect on η_0 and energy.
6. **Deferred: the UHF and xc-kernel paths.** `ssRIURPA` should follow with the stacked-spin convention that `gwfast` already validated (spin factor 1 over $2ov$). EDMET's `rixc` path cannot use the symmetric $D^{1/2}\tilde{M}^{-1/2}D^{1/2}$ form, because $A - B$ is not diagonal. It needs either a non-symmetric Woodbury rational apply (LU solves) on MP, whose interval is not theorem-backed, or $(A-B)^{1/2}$ as a diagonal-plus-low-rank function. Keep it on Clenshaw–Curtis until that is designed. Periodic `kdf` cases also wait, for the reasons `monomial_GW` Milestone 6 records.

3.4 Points to settle before implementation

- **Spin conventions.** `ssRIdRRPA` mixes spatial integrals with spin-orbital outputs (`stack_spin`, `return_spatial`). The factor 4 in $\tilde{M}$ and the split of the spin-diagonal identity term must be checked against the dense `ssRPA` before anything else.
- **Negative CDERIs** (`lov_neg`, from non-PSD metrics in some periodic setups). $A + B$ then takes an indefinite low-rank part, C_j is no longer SPD, and the certified interval argument fails. The HHT path should refuse them at first.
- **Memory for the cache.** Storing the factors costs $N_p N_{\mathrm{aux}}^2$ (about 0.2 GB at $N_{\mathrm{aux}} = 1000$, $N_p = 20$). For very large N_{aux}, rebuild per pole; `gwfast`'s `PRECOMPUTE_S` switch is exactly this trade.

- **Higher moments** are already exact recurrences at $\mathcal{O}(n_t N_{\text{aux}} ov)$ per order. `monomial_GW` Milestone 3 found that its high-order instabilities were in the downstream realisation, not in this recurrence, so no change is proposed there.

4 Local and complementary projections (draft)

Status: first draft for discussion. All costs are operation counts. Nothing here has been implemented or measured. It builds on the pole form of Section 3.1 and on the discussion of local spaces labelled by x.

4.1 Setup

Each label x defines a local occupied and a local virtual space through orthonormal rows of C^x:

$$\begin{aligned} &c_o\ (o_x \times o), \quad c_v\ (v_x \times v), \quad c_o c_o^{\mathsf{T}} = I, \quad c_v c_v^{\mathsf{T}} = I, \\ &P_o = c_o^{\mathsf{T}} c_o, \quad P_v = c_v^{\mathsf{T}} c_v, \quad Q_o = I - P_o, \quad Q_v = I - P_v, \\ &e_o,\ e_v \text{ complement rows}, \quad e_o^{\mathsf{T}} e_o = Q_o, \quad e_v^{\mathsf{T}} e_v = Q_v, \\ &X = c_o \otimes c_v \quad \text{local ph target, } n_x = o_x v_x \text{ rows.} \end{aligned} \tag{18}$$

n_x is $\mathcal{O}(1)$ in system size. There are $\mathcal{O}(N)$ labels for molecules and $\mathcal{O}(1)$ for solids.

The right-hand side of the quantities discussed is a **complement** of the local space. Two complements have come up, and they are different quantities, not two representations of one:

Name	Projector on the right	Blocks kept	Example
ph complement	$\bar{\Pi} = I - P_o \otimes P_v$	everything except local–local, including the mixed local-occ/env-vir and env-occ/local-vir blocks	$\tilde{\eta}^x_{i\bar{a},kc}$ (full MO indices on the right)
strict environment	$Q_o \otimes Q_v$	env–env only	$\eta^x_{\bar{i}\bar{a},\bar{k}\bar{c}}$ (complement barred indices)

Either can be **represented** in the full MO basis (apply the projector) or in the complement basis (rotate with $e_o \otimes e_v$). As George notes, the basis is a choice. Section 4.3 shows that the MO-basis representation is the cheaper one to compute.
To decide: which complement each use needs.

4.2 Shared and per-label factors

Shared factors, built once per mean field, as in Section 3.1:

$$F_j = D(D^2 + s_j)^{-1}, \qquad C_j = I + 4LF_jL^{\mathsf{T}} \ \to\ \text{chol}(C_j), \qquad \eta_0 \approx \sum_j w_j [F_j - 4F_jL^{\mathsf{T}} C_j^{-1} LF_j]. \tag{19}$$

They cost $N_p(N_{\text{aux}}^2 ov + N_{\text{aux}}^3/3)$ and need $N_p N_{\text{aux}}^2$ memory.

Per-label left factors, built for each x and each pole:

$$B_j^x = XF_jL^{\mathsf{T}} \ \ (n_x \times N_{\text{aux}}), \qquad \text{Kronecker: } (o_x\, o\, v + o_x v_x\, v)\, N_{\text{aux}}, \tag{20}$$

$$G_j^x = w_j\, B_j^x C_j^{-1} \ \ (n_x \times N_{\text{aux}}), \qquad \text{one Cholesky solve: } n_x N_{\text{aux}}^2. \tag{21}$$

B_j^x is applied in factored form: reshape F_jL^{T} to (o, v, N_{aux}) and contract c_o, then c_v. It is never formed as a dense $n_x \times ov$ target.

Every quantity below is built from these factors, through one of three kernels.

4.3 Kernel A: one-sided, with a projector on the right

$$X\eta_0 R, \qquad R \in \{\, I,\ \bar{\Pi},\ Q_o \otimes Q_v \,\}. \tag{22}$$

Bare term. It is XD_rR with $D_r = \mathrm{diag}(\sum_j w_j F_j) \approx I$. For $R = I$ it is about X. For either complement $XR = 0$ exactly, so the bare term is $X(D_r - I)R$, which is $\mathcal{O}(\mathrm{tol})$. It can be dropped, leaving only the correlated part, with no large cancellation.

Low-rank term. Sum over the poles *before* projecting:

$$S^x = -4 \sum_j G_j^x\,(LF_j) \quad (n_x \times ov,\quad N_p\, n_x N_{\mathrm{aux}} ov), \qquad X\eta_0 R \approx S^x R. \tag{23}$$

The projector has to come after the pole sum. F_j is diagonal in the MO basis and does not commute with R, so applying R inside the sum would cost one application per pole. Here R is applied once per label:

R	How	Cost per label
$\bar{\Pi}$	$S - (SX^{\mathsf{T}})X$, where $SX^{\mathsf{T}} = -4\sum_j G_j^x (B_j^x)^{\mathsf{T}}$ is the two-sided block (kernel B, case a), so it is already available	$N_p n_x^2 N_{\mathrm{aux}} + {\sim} n_x ov\, o_x$ to expand
$Q_o \otimes Q_v$, MO basis	$S - S(P_o \otimes I) - S(I \otimes P_v) + S(P_o \otimes P_v)$, each a low-rank subtraction	${\sim}2\, n_x ov(o_x + v_x)$
$Q_o \otimes Q_v$, complement basis	rotate S by e_o^{T}, e_v^{T}	$n_x(o_{\mathrm{env}} ov + o_{\mathrm{env}} v_{\mathrm{env}} v)$; larger, so use only if that basis is needed downstream

Cost per label.

$$N_p\Big[(o_x\, o\, v + o_x v_x\, v) N_{\mathrm{aux}} + n_x N_{\mathrm{aux}}^2 + \underbrace{n_x N_{\mathrm{aux}} ov}_{\text{dominant, output-bound}}\Big] + \mathcal{O}\big(n_x ov(o_x + v_x)\big). \tag{24}$$

The output has $n_x ov$ entries, so the $n_x N_{\mathrm{aux}} ov$ back-multiply per pole cannot be avoided while the result is kept open on the right. The only levers on it are N_p and N_{aux} (natural auxiliary functions).

Batching across labels. Stack G_j^x over a batch of labels, so that each pole is one GEMM $(\sum_x n_x \times N_{\mathrm{aux}}) \cdot (N_{\mathrm{aux}} \times ov)$. The flops are the same but the efficiency is better. Consume or store each $S^x R$ before the next batch, because holding all labels is $\mathcal{O}(N^3)$ memory for molecules.

4.4 Kernel B: two-sided, $X\eta_0 Y^{\mathsf{T}}$ with a small Y

$$X\eta_0 Y^{\mathsf{T}} \approx XD_r Y^{\mathsf{T}} - 4\sum_j G_j^x\,(B_j^y)^{\mathsf{T}}, \qquad B_j^y = YF_jL^{\mathsf{T}} \quad (n_Y \times N_{\mathrm{aux}}). \tag{25}$$

Per pole this costs forming B_j^y plus $n_x N_{\mathrm{aux}} n_Y$. The open $n_x \times ov$ block is never built. Kernel B pays off only when $n_Y \ll ov$, or when B_j^y is known without a sweep over ov. That is the case when the right-hand index is contracted with something small *before* anything is stored.

Case	Y	B_j^y	Extra cost per pole	Uses
(a) same space	X	B_j^x (already built)	$n_x^2 N_{\text{aux}}$	screening $T_f \eta_0 T_f^\mathsf{T}$, RPA bath $m_0 T^\mathsf{T}$, the $\bar{\Pi}$ correction in kernel A
(b) RI span	L	$LF_jL^\mathsf{T} = (C_j - I)/4$, free	none: $X\eta_0 L^\mathsf{T} \approx XD_rL^\mathsf{T} - \sum_j w_j B_j^x (I - C_j^{-1})$	GW-type $\eta_0 V$; interactions built from RI vectors
(c) RI span, projected	$L\bar{\Pi}$	$(C_j - I)/4 - (LX^\mathsf{T})B_j^x$	$N_{\text{aux}}^2 n_x$; LX^T once per label	RHS of $\tilde{\eta}^x$ contracted directly with $(kc\|Q)$
(c′) RI span, env–env	$L(Q_o \otimes Q_v)$	$(C_j - I)/4$ minus the P_o, P_v and $P_o \otimes P_v$ corrections, each a factored low-rank product	$\sim(o_x + v_x)ovN_{\text{aux}} + (o_x v + ov_x)N_{\text{aux}}^2$	RHS of $\eta^x_{i\bar{a},\bar{k}\bar{c}}$ contracted with integrals
(d) another label	$X' = c'_o \otimes c'_v$	$B_j^{x'}$, already built for x'	$n_x N_{\text{aux}} n_{x'}$	cross-label couplings, if needed

For (c) and (c′) the bare term XD_rRL^T is again $\mathcal{O}(\text{tol})$ and can be dropped.

Whether kernel B beats kernel A for (c′) depends on the sizes: $N_{\text{aux}}(o_x v + ov_x)$ against $o_x v_x\, ov$ per pole. For small local spaces in a large system, kernel B is expected to win, but this has to be measured.

4.5 Kernel C: fully contracted, $\mathrm{Tr}[X\eta_0 Y^\mathsf{T}]$

$$\mathrm{Tr}\left[X\eta_0 Y^\mathsf{T}\right] \approx \mathrm{Tr}\left[XD_r Y^\mathsf{T}\right] - 4\sum_j w_j \,\mathrm{Tr}\left[C_j^{-1}(B_j^y)^\mathsf{T} B_j^x\right]. \tag{26}$$

This costs $\min(n_X, n_Y)N_{\text{aux}}^2 + N_{\text{aux}}^3$ per pole once the B matrices are built. When $X = Y = L$, it is $-\frac{1}{4}\sum_j w_j \,\mathrm{Tr}[C_j - 2I + C_j^{-1}]$ plus the diagonal term: one Gram and one Cholesky factorisation per pole, with no target contraction. The uses are the cumulant energy, the `"linear"` xc energy correction and `direct_AC_integration` (Section 3.2, item 2).

4.6 Summary by use

Use	Right side	Output	Kernel	Dominant cost per label
$\tilde{\eta}^x_{i\bar{a},kc}$	$\bar{\Pi}$, MO basis	$n_x \times ov$	A	$N_p n_x N_{\text{aux}} ov$
$\eta^x_{i\bar{a},\bar{k}\bar{c}}$, stored	$Q_o \otimes Q_v$	$n_x \times n_{\text{env}}$	A, then rotate if the env basis is needed	$N_p n_x N_{\text{aux}} ov$
either, RHS contracted with L next	$L\bar{\Pi}$ or $L(Q_o \otimes Q_v)$	$n_x \times N_{\text{aux}}$	B (c)/(c′)	$N_p N_{\text{aux}}^2 (o_x v + ov_x)$; no $n_x ov$ term
screening, RPA bath	T_f	$n_x \times n_x$	B (a)	$N_p n_x N_{\text{aux}}^2$ + factored forward
energies	various	scalar	C	N_{aux}^2 per pole

With the shared factors, the totals for kernel A are $\mathcal{O}(N^4)$ for molecules. For solids they are a shared $\mathcal{O}(N^4)$ term plus $\mathcal{O}(N^3)$ per label. Today's route costs about $48N_{\text{aux}}^2 ov + 192\sum_x n_x N_{\text{aux}} ov$ for any of these, and always builds the $n_x \times ov$ block. With the shared factors, kernel A costs about $20N_{\text{aux}}^2 ov + 20\sum_x n_x N_{\text{aux}} ov$, about 2.4× on the shared term and about 10× on the per-label term. Kernels B and C remove the per-label ov sweep where the use allows it.

4.7 Proposed kernel interface

```
# s_j, w_j, F_j, chol(C_j); certified interval
poles = build_pole_factors(D, L, tol)
# B_j^x, G_j^x (Kronecker-factored)
left = local_left_factors(poles, c_o, c_v)
# kernel A
one_sided(poles, left, right=None | "ph_complement" | "env", basis="mo" | "env")
# kernel B; right = left' | "ri" | ("ri", projector)
two_sided(poles, left, right)
# kernel C
trace(poles, left, right)
```

Every function also takes a batch of labels, and each records the per-pole Cholesky residuals and the intermediate shapes, following `monomial_GW` Milestone 2.

4.8 Open questions

1. **Which complement** ($\bar{\Pi}$ or $Q_o \otimes Q_v$) each use needs.
2. **What the right-hand index is contracted with next** in each use. This decides between kernel A and kernel B, and it is the main source of savings beyond the constant factors.
3. **Whether the outputs are stored or consumed** per label. This sets the memory plan for kernel A.
4. **Orthonormality of C^x rows.** The $XR = 0$ cancellation and the cheap projector applications rely on it.
5. **Spin.** Restricted spatial forms are given here, with the factor 4 still to be checked against the dense `ssRPA`. The spin-orbital stacking used by the existing callers has to be mapped onto them.
6. **Dropping the $\mathcal{O}(\mathrm{tol})$ bare term.** It is consistent with the certificate on η_0, but the error bound for the projected quantity should be stated explicitly, for example by Cauchy–Schwarz with $\|X\| = 1$.